x_1	x_2	x_3	x_4	x_5	x_6
$\begin{pmatrix}1\\1\end{pmatrix}$	$\begin{pmatrix}2\\1\end{pmatrix}$	$\begin{pmatrix}1\\2\end{pmatrix}$	$\begin{pmatrix}5\\4\end{pmatrix}$	$\begin{pmatrix}6\\5\end{pmatrix}$	$\begin{pmatrix}6\\6\end{pmatrix}$
C_1	C_1	C_1	C_2	C_2	C_2

$\begin{pmatrix}4\\-2\\-2\end{pmatrix}$ *Initial W* **REPEAT using Step η = 1**

C_1 positive C_2 negative **Zero** is unaccepted

Check classes of $Z_1 = \begin{pmatrix}0\\0\end{pmatrix}$ $Z_2 = \begin{pmatrix}7\\6\end{pmatrix}$ $Z_3 = \begin{pmatrix}3\\2\end{pmatrix}$

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
$\begin{pmatrix}1\\1\\1\end{pmatrix}$	$\begin{pmatrix}1\\2\\1\end{pmatrix}$	$\begin{pmatrix}1\\1\\2\end{pmatrix}$	$\begin{pmatrix}1\\5\\4\end{pmatrix}$	$\begin{pmatrix}1\\6\\5\end{pmatrix}$	$\begin{pmatrix}1\\6\\6\end{pmatrix}$
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 2

No change in W
Stop Iterations

$d(x_1) = \begin{pmatrix}1&1&1\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$

$d(x_2) = \begin{pmatrix}1&2&1\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$

$d(x_3) = \begin{pmatrix}1&1&2\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$

$d(x_4) = \begin{pmatrix}1&5&4\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$

$d(x_5) = \begin{pmatrix}1&6&5\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$

$d(x_6) = \begin{pmatrix}1&6&6\end{pmatrix}\begin{pmatrix}5\\-1\\-1\end{pmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix}5\\-1\\-1\end{pmatrix}$